Project Specification for Unsupervised Learning

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Group 3 - P2

1 Project Title

"Unsupervised Clustering of Hotel Bookings: Segmenting Business and Leisure Travelers Based on Booking Behavior"

2 Problem Statement

Customer segmentation by travel purpose is essential for hotel revenue optimization. Business and leisure travelers exhibit distinct booking patterns: business travelers book last-minute at premium rates while leisure guests plan ahead and respond to discounts. Research shows data-driven segmentation increases average daily rates by 15-25% [1, 2]. This project will apply unsupervised clustering to discover natural customer segments from hotel booking data, identifying patterns that can inform revenue strategies.

3 Problem Framing (RQ1)

RQ1: What is the concrete segmentation question, and why is clustering an appropriate tool?

Segmentation scope

We will segment hotel guests into travel purpose clusters using features like lead time, length of stay, booking day of week, number of adults, children, babies and seasonality to identify if natural business or leisure patterns emerge from booking data.

Strategic value

A data-driven segmentation will reveal which booking characteristics actually distinguish guest segments, enabling hotels to assemble potential future patterns and elaborate better revenue strategies with tailored pricing and marketing approaches.

Methodological justification

Clustering is appropriate because: (1) we have no predefined labels for guest segments in the booking data; (2) it will discover natural groupings empirically from booking patterns rather than forcing data into assumed categories; and (3) it will enable discovery of micro-segments with distinct characteristics that simple business vs. leisure classification would miss.

4 Representation / Preprocessing Choices (RQ2)

RQ2: Which representation, preprocessing, or encoding choices seem justified, and how might they affect similarity and clustering?

Feature selection rationale

We will use 17 numerical features capturing temporal and transactional dimensions (lead time, stay duration, guest composition, booking history, ADR) and 10 categorical features capturing contextual and service dimensions (hotel type, arrival month, meal plan, market segment, distribution channel, room types, deposit type, customer type). Temporal features (lead time, weekend vs. weekday stays) reveal booking timing patterns that distinguish business from leisure travelers. Compositional features (adults, children, babies) capture group structure—leisure travelers book families; business travelers often travel solo. Contextual features (market segment, distribution channel, meal type) directly encode travel purpose signals. Together, these features form a complementary set that jointly capture the multidimensional nature of travel purpose at booking time: when travelers book, how long they stay, who travels with them, and what services they request all cohere to indicate whether a booking is for business or leisure.

Leakage control and dimensionality

Five variables will be excluded to prevent leakage: `is_canceled`, `reservation_status`, and `reservation_status_date` are post-event outcomes unknown at booking time. `agent` and `company` are high-cardinality ID-like fields with no meaningful encoding.

Standardisation and encoding pipeline

Numerical features will be median-imputed (robust to outliers) then standardized (z-score). Categorical features will be mode-imputed, rare categories ($< 1\%$ frequency) grouped into `''Other''`, then one-hot encoded. These choices are justified through exploratory analysis of the dataset: median imputation preserves distribution shape under skew, standardization ensures equal contribution across numerical dimensions in Euclidean space, and rare category grouping reduces noise without losing information. The result will be a $119{,}210 \times 81$ standardised matrix with Euclidean distance in mixed space.

Alignment with RQ1 objectives

These choices will support RQ1 by ensuring clustering uses only booking-time information, preserving temporal and compositional patterns through standardization and encoding, and producing a representation compatible with K-means and Ward hierarchical clustering.

5 Limitations, Risks, and Ethical Issues

Limitations

Clustering-based segmentation depends critically on feature representation and algorithm selection. If booking-time features alone are insufficient to distinguish travel purpose, or if segments are inherently overlapping in feature space, clustering may produce weak or unstable segments. The interpretability of clusters depends on post-hoc profiling using outcome variables; clusters that do not align with intuitive travel-purpose labels may lack actionable meaning for hotel management.

Risks

Clustering outcomes are sensitive to preprocessing choices (imputation strategy, standardization, rare category threshold). Dimensionality reduction during encoding may obscure minority booking patterns. Post-hoc profiling using outcome variables (`is_canceled`, `reservation_status`) risks confirmation bias if patterns are interpreted selectively to match preexisting assumptions about business vs. leisure travelers.

Ethical considerations

Segment-based pricing strategies may raise fairness concerns. Hotels must ensure transparency and avoid algorithmic discrimination, particularly preventing segments from proxying for protected characteristics.

References

[1] Exploring booking intentions through price elasticity of demand in tourism accommodations using large-scale data analytics. ScienceDirect, 2025. https://www.sciencedirect.com/science/article/pii/S2444883425000038

[2] Core Market Segmentation of the Hotel Industry Explained. Prostay, 2025. https://www.prostay.com/blog/market-segmentation-of-hotel-industry/